

课堂复盘与教学分析系统 M1 个人项目报告

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报告说明

本项目为个人独立开发项目，面向课程评审，严格按照给定报告规范撰写；精简原文冗余描述、删除重复赘述，区分**已实现运行**、**部分实现**、**尚未实现**三类功能，逻辑清晰、表述专业，完整覆盖评审要求全部 7 大模块。

一、问题与用户

1.1 目标用户

核心用户为一线授课教师，M1 阶段仅支持单人单节课复盘，不面向学校管理者、区域教研等组织管理场景；兼容线下课堂、直播网课、录播课三类课堂视频素材。

1.2 真实使用场景与现有痛点

教师课后需回看课堂视频，定位提问、学生作答、教师反馈、讲授时长等教学片段，传统复盘存在 6 大核心痛点：

- 1. 整段课堂视频时长久，人工拖拽定位片段效率极低；
- 2. 通用语音识别输出文本碎片化，无法直接用于教学分析；
- 3. 逐字稿、视频片段、复盘结论无绑定关系，核对成本高；
- 4. 通用大模型仅依靠局部文本易生成无依据过度推断；
- 5. 线下、直播、录播课可采集信息不同，无法套用同一套分析逻辑；
- 6. 教师需要绑定课堂实录的可落地改进建议，而非空泛评价。

1.3 项目目标、完整流程与 M1 成功标准

完整闭环流程

上传课堂视频→生成带时间戳逐字稿→教师人工修订→计算课堂指标→生成候选教学证据→教师审核（接受 / 修改 / 驳回）→基于确认内容输出复盘报告

M1 成功验收标准

- 1. 视频稳定上传持久化存储；
- 2. 调用真实 ASR 生成带时间点逐字稿，支持视频精准跳转；

- 3. 教师可修正识别错误，修订文本自动流入下游分析；
- 4. 教学证据绑定原文与视频时间轴；
- 5. 报告仅采用教师人工确认后的分析内容。

1.4 M1 范围收敛原因

项目单人独立开发，开发周期、云服务预算、测试素材、工程人力全部受限。若首期同步开发多用户后台、画面视觉识别、跨课对比、知识图谱等模块，会导致核心复盘主流程无法落地。

M1 仅聚焦**单节课完整处理链路**，优先验证素材接入、逐字稿编辑、指标证据溯源、复盘报告生成四大核心能力，同时为 M2 跨课对比、M3 多课程管理预留架构扩展基础。

二、方案与交互

2.1 核心业务流程

教师创建课堂→选择课堂类型→上传视频→云端存储音视频资源→后台异步任务启动→调用 ASR 生成时间戳逐字稿→Transcript Agent 结构化整理课堂文本→教师修订确认逐字稿→指标引擎计算课堂数据→Teaching Evidence Agent 生成教学证据→教师审核证据→输出复盘报告

当前仅提供结构化操作工作区，AI 对话交互区未落地，暂不支持自然语言指令驱动分析、追问证据依据。

2.2 工作区与规划 AI 对话区分职责

- 1. **现有结构化工作区（已实现）**：承载视频上传播放、逐字稿编辑、片段翻译、指标查看、证据复核、报告预览等实体操作，以播放器、时间轴、编辑面板、证据卡片为核心交互载体；
- 2. **规划 AI 对话区（未实现）**：负责流程引导、状态查询、证据溯源解释、自定义片段分析、异常报错指引；
- 3. 二者分工边界：AI 对话做引导询问，视频校对、文本修改、证据审核、报告确认仍在结构化工作区完成。

2.3 系统关键状态与两个人工确认节点

底层任务状态枚举

created（创建）→queued（排队）→running（处理中）→waiting_for_human（等待人工）→completed（完成）→failed（失败）→cancelled（取消）

前端转换为教师易懂流程：创建课堂→上传素材→生成逐字稿→校订文本→证据审核→人工复核→生成报告

强制人工确认节点（流程断点设计）

- 1. **逐字稿校订节点**：ASR 文本生成后流程暂停，教师完成错别字、发言人修正并确认，系统才启动指标与证据分析；

2. 证据复核节点：候选证据全部生成后，教师逐条审核，仅接受 / 修改后采纳的证据会写入最终报告。

2.4 异常处理现状与后续完善方向

当前仅能识别基础处理异常，缺少标准化错误提示、自动重试、故障定位能力；常见故障场景：视频上传失败、ASR 接口调用异常、发言人识别错乱、后台任务中断、报告生成失败。
后续补充：标准化错误状态、面向教师通俗报错文案、技术日志、重试机制、故障步骤快速恢复入口。

2.5 核心高价值交互设计

逐字稿 / 教学证据双向跳转视频时间轴 + 修订驱动重分析

教师点击任意逐字段落、任意证据卡片，直接定位对应视频片段；教师修改识别文本后，系统自动基于修订后内容重新计算指标、生成证据。
该设计打通视频、文本、指标、分析结论统一时间基准，解决传统复盘 “视频与文字脱节、结论无实录支撑” 的痛点，提升复盘准确度与效率。

三、实现与技术

3.1 整体技术设计三大原则

- 1. 快速闭环验证：单人开发优先降低前后端协同成本，快速打通端到端可用流程；
- 2. 数据全链路可追溯：视频、逐字稿、时间戳、指标、证据、人工审核记录强关联，采用结构化数据库约束数据关系；
- 3. 架构可扩展：适配 M2 两轮课堂对比、M3 多课程管理、RAG / 记忆模块新增需求，无需大规模重构底层代码。

3.2 工程整体架构（Monorepo）

采用 pnpm Workspace + Turborepo 统一管理代码仓，拆分应用层与公共包：

代码块

- 1 apps/ web前端、api业务接口、worker异步任务
- 2 packages/ 数据库层、第三方服务封装、智能Agent、指标计算、数据类型、接口契约

优势：全栈共用一套 TypeScript 类型定义、接口规范、状态枚举，规避单人开发前后端字段不一致问题，支持模块独立部署替换。

3.3 分层技术选型及用途（仅罗列实际落地技术）

1. 前端：Next.js + React + TypeScript

实现课堂创建、视频分片上传、进度展示、时间轴播放器、逐字稿编辑、证据审核、报告页面；组件化拆分交互面板，TS 保障类型统一。

2. 后端 API：NestJS + TypeScript

管理课堂元数据、生成云存储上传凭证、持久化人工修订 / 审核记录、触发报告生成；模块化拆分课堂、上传、逐字稿、证据业务。

3. 异步后台 Worker（独立进程）：TypeScript

异步处理耗时任务：视频音频解析、ASR 调用、逐字稿结构化、指标计算、Agent 证据生成；隔离长耗时任务，避免前端接口阻塞，支持任务排队、独立扩缩资源。

4. 数据库：Supabase PostgreSQL

存储全链路结构化关联数据：课堂信息、音视频元数据、带时间戳逐字稿、修订记录、指标、教学证据、教师审核结果、报告文本；依托关系型数据库实现数据溯源，预留向量字段支持后续 RAG。

5. 对象存储：Cloudflare R2

存储原始课堂视频、ASR 提取音频、导出报告文件；浏览器直传签名地址，避免大文件经过 API 中转，分离存储与计算资源。

6. 外部第三方服务

- 阿里云 DashScope ASR：音频转带时间戳逐字稿，使用真实商用语音服务而非模拟数据；
- 通用大语言模型：Transcript 文本整理、课堂分段标题、教学证据生成、双语翻译、优化教学建议话术；通过统一 Provider 层封装调用，解耦业务与模型厂商。

7. 两大自研智能 Agent 模块

- Transcript Agent：标准化逐字稿双投影架构（展示投影供教师阅读编辑、分析投影供指标计算），基于教学活动语义拆分课堂段落；
- Teaching Evidence Agent：依据修订文本、课堂类型、互动链生成教学证据，内置**可分析能力矩阵**区分线下 / 直播 / 录播分析边界，输出附带不确定性说明、可落地教学建议。

8. 指标引擎 Metrics Engine：纯程序逻辑计算语速、讲授时长、提问等待时长、师生互动链、信息密度，不依赖大模型，保证数值稳定可控。

9. 云端部署：Google Cloud Build+Cloud Run（分离部署前端、API、Worker）、Google Secret Manager 保管密钥；无自建服务器，按需弹性运行。

四、完成范围与限制

4.1 已真实实现、全链路可运行功能

覆盖线下课堂、录播课完整处理闭环，所有功能接入真实云端服务、可生成完整复盘报告：

1. 课堂创建、分类选择、视频分片上传 + 进度展示、音视频持久化；
2. 商用 ASR 生成带时间戳逐字稿、逐字稿编辑保存、英文片段即时翻译；

3. Transcript Agent 语义分段、生成教学段落标题、双投影数据结构；
4. 基础课堂指标计算、「提问 - 等待 - 作答 - 反馈」互动链分析、信息密度评估；
5. Teaching Evidence Agent 生成多类型证据卡片、证据绑定原文与视频时间点；
6. 证据三级审核（接受 / 修改采纳 / 驳回）、仅采纳内容生成复盘报告；
7. 能力矩阵区分三类课堂分析边界、低置信片段增加不确定性提示。

4.2 部分实现（核心逻辑落地，细分能力未完善）

1. **Transcript Agent**：逐字稿生成、编辑、下游重分析全部完成；发言人识别受教室噪声、重叠人声限制，无法稳定区分单人作答 / 全班齐答，缺少发言人手动编辑入口、完整版本记录、逐字稿质量评测工具。
2. **Teaching Evidence Agent**：证据生成、溯源、审核流程可用；五级学习检查、课堂细分结构、复习课 / 试卷讲评课专属分析规则仅完成设计，未全量接入。
3. **报告生成**：可输出基础确认内容汇总；独立 Report Agent、多格式导出、报告版本管理、生成校验机制未实现。

4.3 尚未实现（仅设计规划，无真实代码落地）

1. M1 补齐类：AI 对话交互区、完整运行日志、自动化失败重试、课堂节奏曲线、PDF 报告导出、直播聊天 CSV 导入、课件关联、课堂截图证据、系统化评测体系；
2. M2 阶段：两轮课堂改进对比、轻量化 Memory 记忆、跨课 RAG 检索；
3. M3 阶段：多课程管理、长期教学趋势汇总、全量教学知识库 RAG；
4. 扩展功能：课堂知识点脑图、师生问答关系图谱。

4.4 项目运行上限与受限根因

1. 当前运行上限：单节线下 / 录播课堂可完整跑完从视频上传到报告生成全流程；经两段真实课堂素材完整验证；
2. 核心限制原因：单人独立开发、周期有限，无专职测试 / 算法 / 前端协作；云模型与视频存储存在调用成本；缺少人工标注课堂数据集；多云服务联调工作量大；发言人识别受教室录音硬件、重叠语音客观约束；观测、评测系统需额外大量工程开发。

4.5 分阶段后续补齐计划

1. **M1 收尾阶段**：补齐 AI 对话模块、独立 Report Agent、全链路观测日志、自动化重试、课堂节奏曲线、PDF 导出、发言人手动修正功能，搭建标准化离线评测数据集；
2. **M2 迭代**：开发课堂改进对比模块，增加 Memory 存储历史教学问题与改进计划，轻量 RAG 匹配两轮课堂相似教学片段；
3. **M3 迭代**：多课程管理、长期教学趋势汇总、全量教学资料 RAG 检索；

4. 可选扩展：课堂知识点关系图谱，所有图谱节点绑定视频时间戳，支持人工修正。

五、测试记录

5.1 整体测试覆盖现状

完成录播课、线下课堂两类素材完整端到端流程测试，均可正常输出报告；未开展并发、压力、多浏览器、大规模素材、故障恢复系统化测试，仅能证明主流程可用，未达到生产级稳定标准。

5.2 完整流程测试案例：线下课堂视频测试

1. 测试目标：验证师生语音识别、互动链分析、证据溯源、人工修订重分析全链路；
2. 测试流程：创建线下课堂→上传实拍教室视频→ASR 生成师生逐字稿→人工修正识别错误→确认文本后计算提问等待时长、师生互动链→生成课堂互动类证据→逐条审核证据→生成复盘报告；
3. 测试结果：互动链可正常参与分析，证据均可跳转对应视频片段；**核心缺陷**：多人重叠发言、全班齐答场景发言人标签识别错乱，干扰作答次数、等待时长计算置信度。

5.3 典型失败场景：发言人识别不准确

1. 故障现象：ASR 说话人分离将教师识别为学生、单人学生拆分为多个发言人、齐答统一标记为单人发言；
2. 产生根源：教室收音设备距离教师更近、学生音量弱、室内回声、多人语音重叠、商用 ASR 分人能力有限，无额外身份标注数据辅助；
3. 连锁影响：提问 - 作答匹配关系错乱、学生作答统计失真、教学证据置信度下降；
4. 当前缓解方案：不单一依赖发言人标签，结合上下文教学行为辅助判断；低置信片段添加不确定性提示；证据全部附带原文与视频时间，方便教师人工核对；教师可修改逐字稿降低分析偏差；
5. 后续修复计划：新增发言人角色手动编辑功能，区分齐答 / 单人作答逻辑，修正发言人后自动重算全部指标与证据，扩充对应测试片段。

六、AI 协作说明

6.1 产品运行阶段内置 AI 能力

Transcript Agent、Teaching Evidence Agent、商用 ASR、通用大语言模型；指标计算、流程状态、数据校验均为原生程序代码，与语义生成模块解耦，保障数值稳定性。

6.2 AI 辅助完成的开发工作

1. 调研阶段：课堂复盘行业产品调研、ASR / 对象存储 / 异步任务技术方案对比、Agent/RAG/Memory 适用场景梳理；
2. 方案设计：M1/M2/M3 阶段范围划分、系统 Workflow 流程、逐字稿双投影数据结构、教学证据数据模型、不同课堂类型分析边界设计；

3. 代码开发：生成基础 TS 类型定义、接口契约、页面 / API/Worker 基础代码、第三方云服务适配模板、批量重构脚本；
4. 调试测试：定位 TS 编译报错、核对前后端接口字段一致性、排查云密钥配置、生成异常测试场景、梳理发言人识别缺陷案例、校验模型输出是否超出课堂素材范围。

6.3 全部由人工独立判断、修改定稿内容

AI 输出的代码、方案、文案均经过人工审核修改，项目最终交付物所有核心决策由本人确定：

项目范围边界、分层技术架构、逐字稿双投影核心设计、教学证据能力矩阵、修订驱动重分析逻辑、三类课堂差异化分析规则、教学建议表述风格、代码编译运行校验、云端服务联调、测试结果有效性判定。

系统最终功能以人工调试后的可运行代码、两段真实素材完整测试结果为准。

七、个人完成内容

7.1 项目全部职责（纯个人独立开发，无协作成员）

需求分析、产品流程设计、前后端全栈开发、异步 Worker 开发、数据库模型设计、云存储 / ASR / 大模型第三方接入、双 Agent 模块设计实现、指标引擎开发、证据审核流程、报告生成逻辑、云端部署、全流程测试、项目技术文档撰写。

7.2 本人原创核心设计与落地成果

1. **逐字稿双投影结构**：单份原始文本拆分阅读、分析双层存储结构，精准区分教师、单人学生、集体发言场景，同时适配人工校对操作与后台指标自动化运算。
2. **课型差异化分析矩阵**：统一课程分类标准，针对线下、直播、录播三类课堂设置独立分析权限，杜绝无素材支撑的无效教学推断，保障分析客观性。
3. **师生互动完整建模**：搭建提问-等待-回答-反馈全流程分析链路，划分6类课堂回应模式，量化学生思考时长，摒弃传统粗放的提问数量统计方式。
4. **五级课堂检查分级**：依据课堂提问的认知深度，划分五级学情检查标准，对低效的集体口头确认标注低可信度，引导深度学情检测，避免学情误判。
5. **教学话术过滤机制**：智能隔离课堂管理、设备调试等非教学话术，仅统计具备学科认知考察价值的提问，从源头保障指标计算精准度。
6. **多课堂场景自适应**：可自动识别新授课、试卷讲评、复习、刷题训练四类课堂场景，适配不同课堂的专属评价逻辑，贴合基础教育教学场景。
7. **双层证据输出设计**：区分机器底层量化运算层与教师可视化展示层，对外输出温和、具象、绑定课堂视频片段的优化建议，无空泛、否定式评判话术。
8. **证据溯源校验体系**：所有教学证据绑定原文文本与视频时间轴，自动拦截越界分析、低置信度结论，对存疑内容标注说明，实现全分析链路可追溯。

7.3 开发过程核心难点与解决思路

- 1. 大视频上传状态同步难题：采用浏览器直传 R2 + 数据库异步记录上传进度，规避 API 中转大文件；
- 2. ASR 碎片化文本无法用于教学分析：自研 Transcript Agent 语义重组，构建双投影结构化数据；
- 3. 发言人识别误差干扰分析：多维度上下文辅助校验，增加人工修正入口、低置信度提示；
- 4. 模型输出易脱离课堂实录：增加证据来源、时间戳、原文三重校验机制；
- 5. 多云端服务配置复杂：统一 Secret 管理、封装第三方 Provider 层统一调用规范。

7.4 项目最终交付成果

搭建一套可端到端运行的单课堂 AI 复盘系统，使用线下、录播两类真实课堂视频完成完整闭环验证；系统可自动生成带溯源证据的教学复盘报告，解决教师传统复盘效率低、分析无依据的核心痛点；当前核心流程验证完成，细分精细化功能受个人资源、周期限制分阶段迭代补齐。

总结

本项目围绕一线教师课后复盘真实痛点，单人独立完成 M1 单课堂完整 AI 分析闭环，采用分层可扩展技术架构，通过逐字稿时间轴溯源、人工修订重分析、课堂类型差异化分析三大核心创意提升复盘可信度与使用效率。报告清晰区分已落地、部分落地、待开发功能，客观说明资源与技术约束，并给出分阶段迭代计划；所有功能均经过真实课堂素材测试，完整验证产品核心业务价值。

Classroom Reflection and Teaching Analysis System M1 Personal Project Report

Report Description

This is an independently developed personal project for course evaluation. The report is strictly compiled in accordance with official specifications, with redundant descriptions streamlined and repeated content removed. It clearly distinguishes fully implemented, partially implemented, and unimplemented functions, maintains rigorous logic and professional expression, and covers all seven core evaluation modules.

1. Problem & User Analysis

1.1 Target Users

The core users are front-line teachers. The M1 phase only supports single-lesson personal classroom reflection and does not serve school administrators or regional teaching research scenarios. The system is compatible with three types of classroom video materials: offline in-person classes, live online classes, and pre-recorded online classes.

1.2 Application Scenarios & Existing Pain Points

Teachers need to review full-class videos after class to locate key teaching segments such as questioning, student responses, teacher feedback, and lecture duration. Traditional manual classroom reflection faces six core pain points:

1. Full-length classroom videos are time-consuming, and manually locating effective segments is extremely inefficient.
2. Raw ASR speech recognition outputs fragmented text that cannot be directly applied to teaching analysis.
3. Transcripts, video clips, and reflection conclusions are disconnected, resulting in high verification costs.
4. General large language models tend to generate unsupported and excessive inferences based on limited text content.
5. Offline, live, and pre-recorded classes have different available data dimensions and cannot adopt a unified analysis logic.

6. Teachers require practical, video-based optimization suggestions rather than empty generalized evaluations.

1.3 Project Objectives, Full Workflow & M1 Acceptance Criteria

Closed-Loop Workflow

Upload classroom video → Generate timestamped transcript → Manual transcript revision → Calculate classroom indicators → Generate candidate teaching evidence → Teacher review (Accept / Modify / Reject) → Output final reflection report based on verified content

M1 Success & Acceptance Standards

1. Stable video upload and persistent cloud storage.
2. Adopt commercial ASR to generate timestamped transcripts and support precise video positioning.
3. Allow teachers to correct recognition errors, with revised text automatically applied to downstream analysis.
4. Bind all teaching evidence to original transcript content and video timelines.
5. The final report only adopts manually verified and confirmed analysis results.

1.4 M1 Scope Limitation Reasons

The project is independently developed with limited development cycles, cloud service budgets, test materials, and manpower. Simultaneous development of multi-user backend, visual recognition, cross-lesson comparison, and knowledge graph modules in the initial phase would block the implementation of the core reflection workflow. The M1 phase focuses exclusively on the **end-to-end processing pipeline for single lessons**, prioritizing the verification of material access, transcript editing, indicator and evidence traceability, and report generation. Meanwhile, it reserves scalable architecture support for M2 cross-lesson comparison and M3 multi-course management.

2. Solution & Interaction Design

2.1 Core Business Workflow

Teacher creates classroom project → Selects classroom type → Uploads video → Cloud storage of audio and video resources → Background asynchronous task initialization → ASR timestamped transcript generation → Transcript Agent structured text optimization → Teacher transcript confirmation → Metric engine calculation → Teaching Evidence Agent evidence generation → Teacher evidence review → Final report output. Currently, only the structured

workspace is implemented; the AI dialogue module is undeveloped, and natural language-driven analysis and evidence inquiry are not yet supported.

2.2 Role Division: Structured Workspace vs. Planned AI Dialogue

1. **Implemented Structured Workspace:** Supports video playback, transcript editing, subtitle translation, indicator viewing, evidence review, and report preview. Core interactions rely on video players, timelines, edit panels, and evidence cards.
2. **Unimplemented AI Dialogue Panel:** Planned functions include process guidance, status inquiry, evidence traceability explanation, custom segment analysis, and error prompt guidance.
3. **Responsibility Boundary:** The AI dialogue module undertakes auxiliary guidance, while core operations including video proofreading, text modification, evidence review, and report confirmation are completed in the structured workspace.

2.3 System Status Mechanism & Dual Manual Confirmation Nodes

Backend Task Status Enumeration

Created → Queued → Running → Waiting_for_human → Completed → Failed → Cancelled.

Frontend user-friendly process: Create project → Upload materials → Generate transcript → Revise text → Review evidence → Manual verification → Generate report

Mandatory Manual Checkpoints (Process Breakpoint Design)

1. **Transcript Revision Checkpoint:** The workflow pauses after raw ASR text generation. Indicator calculation and evidence analysis only start after teachers correct text errors and speaker labels and confirm the transcript.
2. **Evidence Review Checkpoint:** After candidate evidence is generated, teachers review each item one by one. Only accepted or modified evidence is written into the final report.

2.4 Exception Handling Status & Future Optimization

The current system only supports basic exception identification and lacks standardized error prompts, automatic retry, and fault location mechanisms. Common failures include video upload errors, ASR API exceptions, speaker recognition disorder, background task interruption, and report generation failure. Future optimizations will cover standardized error statuses, teacher-friendly prompt texts, technical logs, automatic retry strategies, and rapid recovery entries.

2.5 Core High-Value Interaction Design

Dual-way Jump Between Transcript/Evidence and Video Timeline + Revision-Driven Re-analysis. Teachers can click any transcript paragraph or evidence card to precisely locate the

corresponding video segment. Any manual modification to the transcript will automatically trigger recalculation of classroom indicators and regeneration of relevant evidence. This design unifies the time benchmark of videos, texts, and analysis conclusions, solves the disconnection between video footage and analytical results in traditional reflection, and greatly improves the accuracy and efficiency of classroom review.

3. Implementation & Technology Stack

3.1 Three Core Technical Design Principles

1. **Rapid Closed-Loop Verification:** Simplify front-end and back-end collaboration logic to quickly implement a usable end-to-end workflow.
2. **Full-Link Data Traceability:** Establish strong correlations among videos, transcripts, timestamps, indicators, evidence, and manual review records through structured databases.
3. **Scalable Architecture:** Adapt to M2 cross-lesson comparison, M3 multi-course management, and RAG/memory module expansion without large-scale underlying reconstruction.

3.2 Overall Monorepo Engineering Architecture

Adopt pnpm Workspace + Turborepo for unified repository management, divided into application layers and public packages:

apps/ (Frontend Web, API Business Interface, Asynchronous Worker)

packages/ (Database Layer, Third-Party Service Encapsulation, Intelligent Agent, Indicator Calculation, Data Type & Interface Contract)

Advantages: The full stack shares unified TypeScript type definitions, interface specifications, and status enumerations, avoiding field inconsistency in independent development and supporting independent module deployment and replacement.

3.3 Technical Selection & Practical Implementation

1. **Frontend:** Next.js + React + TypeScript. Implements classroom project creation, segmented video upload, progress display, timeline player, transcript editing, evidence review, and report preview page. Componentized panel design and TypeScript ensure type consistency.
2. **Backend API:** NestJS + TypeScript. Manages classroom metadata, generates cloud storage upload credentials, persists manual revision and review records, and triggers report generation. Modularly splits business logic for classroom management, file upload, transcript processing, and evidence analysis.
3. **Asynchronous Worker (Independent Process):** TypeScript. Handles time-consuming tasks including video audio parsing, ASR invocation, transcript structuring, indicator calculation, and Agent evidence generation. Isolates long-time tasks to prevent front-end interface blocking and support task queuing and elastic scaling.

4. **Database:** Supabase PostgreSQL. Stores full-link structured data: classroom information, audio and video metadata, timestamped transcripts, revision records, classroom indicators, teaching evidence, teacher review results, and report content. The relational database ensures data traceability and reserves vector fields for subsequent RAG iteration.
5. **Object Storage:** Cloudflare R2. Stores original classroom videos, ASR-extracted audio files, and exported reports. Adopts browser direct upload with signed URLs to avoid API forwarding of large files and separate storage and computing resources.
6. **Third-Party External Services**
 - Alibaba DashScope ASR: Converts audio to timestamped transcripts with commercial-grade speech recognition instead of simulated data.
 - General Large Language Model: Responsible for transcript optimization, paragraph title generation, teaching evidence creation, bilingual translation, and instructional suggestion polishing. Unified Provider layer decouples business logic and model vendors.
7. **Two Self-Developed Intelligent Agents**
 - Transcript Agent: Implements dual-projection transcript architecture (display projection for teacher editing & reading, analysis projection for indicator calculation) and semantic paragraph segmentation based on teaching activities.
 - Teaching Evidence Agent: Generates teaching evidence based on revised transcripts, classroom types, and interaction chains. Built-in capability matrix distinguishes analysis boundaries for offline, live, and pre-recorded classes, outputting objective evidence and practical optimization suggestions with uncertainty notes.
8. **Metrics Engine:** Adopts pure program logic to calculate speech speed, lecture duration, question waiting time, teacher-student interaction chains, and information density. Model-free calculation ensures stable and accurate numerical results.
9. **Cloud Deployment:** Google Cloud Build + Cloud Run (separated deployment for frontend, API, and Worker) + Google Secret Manager for key management. Serverless elastic operation without self-built physical servers.

4. Functional Completion & Limitations

4.1 Fully Implemented & Operational Functions

Complete end-to-end closed-loop processing for offline and pre-recorded classes with real cloud service integration and full report generation:

1. Classroom creation, category selection, segmented video upload with progress display, and audio-video persistence.

2. Commercial ASR timestamped transcript generation, transcript editing, and real-time English translation for text segments.
3. Transcript Agent semantic segmentation, teaching paragraph title generation, and dual-projection data structure implementation.
4. Basic classroom indicator calculation, complete "question-wait-response-feedback" interaction chain analysis, and information density evaluation.
5. Teaching Evidence Agent multi-type evidence card generation with original text and video timestamp binding.
6. Three-level evidence review mechanism (Accept / Modify / Reject), with reports generated only from verified evidence.
7. Classroom type capability matrix for analysis boundary control and uncertainty prompt for low-confidence segments.

4.2 Partially Implemented Functions (Core Logic Completed, Details Undeveloped)

1. **Transcript Agent:** Transcript generation, editing, and downstream re-analysis are fully implemented. Speaker recognition is unstable for overlapping student voices and noisy classroom environments, lacking manual speaker editing entries, complete version records, and transcript quality evaluation tools.
2. **Teaching Evidence Agent:** Evidence generation, traceability, and review workflows are available. The five-level learning inspection mechanism and exclusive analysis rules for review and exercise-commentary lessons are only designed but not fully integrated.
3. **Report Generation:** Basic verified content summarization is available. Independent Report Agent, multi-format export, report version management, and generation verification mechanisms are undeveloped.

4.3 Unimplemented Functions (Designed but Not Coded)

1. M1 Supplementary Functions: AI dialogue panel, complete operation logs, automatic failure retry, classroom rhythm curve, PDF report export, live chat CSV import, courseware association, screenshot evidence, and systematic evaluation system.
2. M2 Iteration: Dual-lesson teaching improvement comparison, lightweight memory module, and cross-lesson RAG retrieval.
3. M3 Iteration: Multi-course management, long-term teaching trend summary, and full-knowledge-base RAG retrieval.
4. Extended Functions: Classroom knowledge mind map and teacher-student Q&A relationship graph with timestamp binding.

4.4 Operation Boundaries & Core Limitations

1. **Current Operation Boundary:** Supports complete end-to-end processing for single offline and pre-recorded lessons, fully verified with real classroom video materials.
2. **Core Limitations:** Independent single-person development with limited cycles and cloud budgets; no professional testing or algorithm collaboration; lack of labeled classroom datasets; objective constraints of classroom audio pickup and overlapping voices; high engineering costs for observation and evaluation systems.

4.5 Phased Future Optimization Plan

1. **M1 Completion:** Develop AI dialogue module, independent Report Agent, full-link logs, automatic retry mechanism, classroom rhythm curve, PDF export, and manual speaker editing functions; build standardized offline evaluation datasets.
2. **M2 Iteration:** Develop dual-lesson comparison module, add memory storage for teaching problems and improvement plans, and apply lightweight RAG to match similar teaching segments across lessons.
3. **M3 Iteration:** Implement multi-course management, long-term teaching trend statistics, and full teaching knowledge base RAG retrieval.
4. **Optional Expansion:** Build classroom knowledge relationship graphs with editable timestamp nodes for manual correction.

5. Testing Records

5.1 Overall Test Coverage

Complete end-to-end workflow tests for offline and pre-recorded classroom materials with valid report outputs. No systematic tests for concurrency, pressure, multi-browser compatibility, large-scale materials, and fault recovery. The current system meets basic availability standards but not production-level stability.

5.2 Full Workflow Test Case: Offline Classroom Video

1. **Test Objective:** Verify full-link effectiveness including speech recognition, interaction chain analysis, evidence traceability, and manual revision-driven re-analysis.
2. **Test Process:** Create offline classroom project → Upload real classroom video → Generate ASR transcript → Manually correct recognition errors → Confirm transcript → Calculate question waiting time and teacher-student interaction indicators → Generate interactive teaching evidence → Complete manual review → Output final report.

3. **Test Result:** The interaction chain analysis and video-evidence jump function work normally.
Core Defect: Speaker labeling is disordered under overlapping student speeches and group answering scenarios, reducing the accuracy of response statistics and waiting time calculation.

5.3 Typical Failure Case: Inaccurate Speaker Recognition

1. **Failure Phenomenon:** ASR speaker separation misidentifies teachers as students, splits single student speeches into multiple speakers, and marks group answers as individual responses.
2. **Root Cause:** Uneven classroom audio pickup (closer to teachers), low student volume, indoor echo, overlapping speeches, and inherent limitations of commercial ASR without identity annotation data.
3. **Chain Impact:** Mismatched question-response relationships, distorted student participation statistics, and reduced teaching evidence confidence.
4. **Current Mitigation:** Adopt multi-dimensional contextual teaching behavior judgment instead of relying solely on speaker labels; add uncertainty prompts for low-confidence segments; bind all evidence with original text and timestamps for manual verification; support transcript modification to reduce analysis deviation.
5. **Future Fix Plan:** Develop manual speaker editing functions, optimize group/single answer classification logic, support full indicator and evidence recalculation after speaker correction, and expand test datasets.

6. AI Collaboration Explanation

6.1 Built-In AI Capabilities in System Operation

Transcript Agent, Teaching Evidence Agent, commercial ASR, and general large language models are integrated into the system. Indicator calculation, process status management, and data verification are implemented through native code and decoupled from semantic generation modules to ensure numerical stability.

6.2 AI-Assisted Development Work

1. **Research Phase:** Industry product research, technical solution comparison for ASR/object storage/asynchronous tasks, and scenario sorting for Agent/RAG/Memory.
2. **Design Phase:** M1/M2/M3 scope division, system workflow design, dual-projection transcript structure, teaching evidence data model, and multi-scene analysis boundary formulation.
3. **Development Phase:** Generation of TS type definitions, interface contracts, basic page/API/Worker code, third-party service adaptation templates, and batch reconstruction scripts.

4. **Debug & Test Phase:** TS error troubleshooting, front-backend interface consistency verification, cloud key configuration inspection, abnormal scenario generation, defect sorting, and model output compliance verification.

6.3 Fully Manual Independent Judgment & Finalization

All AI-generated code, solutions, and copywriting have been manually reviewed and revised. All core project decisions are independently confirmed by the developer, including project scope, hierarchical technical architecture, dual-projection transcript design, teaching evidence capability matrix, revision-driven re-analysis logic, multi-scene differentiation rules, teaching suggestion styles, code verification, cloud deployment debugging, and test result validation. The final system deliverables are based on manually optimized runnable code and verified real-scene test results.

7. Personal Contribution & Core Achievements

7.1 Full Personal Responsibilities (100% Independent Development)

Responsible for full-process work including requirement analysis, product workflow design, full-stack development, asynchronous Worker development, database schema design, third-party cloud service integration (Storage/ASR/LLM), dual-Agent logic implementation, metric engine development, evidence review workflow, report generation logic, cloud deployment, full-link testing, and technical documentation writing.

7.2 Original Core Designs & Practical Achievements

1. **Dual-Projection Transcript Architecture:** Divides a single original text into display layer and analysis layer, accurately distinguishing teacher, individual student, and group speech scenarios, adapting to both manual editing and automatic indicator calculation.
2. **Lesson-Type Differentiated Analysis Matrix:** Establishes unified classroom classification standards and independent analysis permissions for offline, live, and pre-recorded classes, eliminating unsupported invalid inferences and ensuring objective analysis results.
3. **Complete Teacher-Student Interaction Modeling:** Builds a full "question-wait-response-feedback" analysis chain, defines 6 types of classroom response modes, quantifies student thinking time, and replaces traditional rough question-counting evaluation methods.
4. **Five-Level Classroom Inspection Grading System:** Classifies classroom questioning by cognitive depth, marks low-confidence for inefficient group oral confirmation, and guides in-depth learning detection to avoid misjudging student mastery.
5. **Teaching Speech Filtering Mechanism:** Intelligently isolates non-teaching speeches such as classroom management and equipment debugging, only retaining discipline-specific cognitive questions to ensure indicator accuracy.

6. **Multi-Scenario Adaptive Analysis:** Automatically identifies four mainstream classroom types (new lesson, exercise commentary, review, and drill training) and matches exclusive evaluation logic suitable for basic education scenarios.
7. **Dual-Layer Evidence Output Mechanism:** Separates underlying machine quantitative calculation and teacher-oriented visualization display, outputting mild, specific, video-bound optimization suggestions without empty or negative evaluations.
8. **Evidence Traceability & Verification System:** Binds all teaching evidence to original text and video timelines, automatically intercepts cross-boundary and low-confidence analysis, marks uncertain content, and realizes full-link traceability.

7.3 Core Development Difficulties & Solutions

1. **Large Video Upload Synchronization Problem:** Adopts Cloudflare R2 direct browser upload + asynchronous database progress recording to avoid API forwarding of large files.
2. **Fragmented ASR Text Inapplicability:** Develops Transcript Agent semantic reorganization and dual-projection structured data construction for teaching-oriented analysis.
3. **Speaker Recognition Deviation Interference:** Introduces multi-dimensional contextual auxiliary verification, manual revision entry, and low-confidence uncertainty prompts.
4. **Model Analysis Detached from Reality:** Implements triple verification of evidence source, timestamp, and original text to restrict model arbitrary inference.
5. **Complex Multi-Cloud Service Configuration:** Unifies secret management and encapsulates third-party Provider layers to standardize external service invocation.

7.4 Final Project Deliverables

Independently built an end-to-end operational AI single-lesson classroom reflection system, fully verified with real offline and pre-recorded classroom videos. The system automatically generates traceable teaching reflection reports, solving the low efficiency and unsupported analysis pain points of traditional manual classroom review. Core business workflows are fully verified, with refined functions to be iterated in phases due to individual development resource and cycle constraints.

Conclusion

Centering on the real post-class reflection pain points of front-line teachers, this project independently completes the full AI analysis closed-loop for single lessons in the M1 phase. Adopting a scalable layered architecture, it improves reflection accuracy and efficiency through three core innovations: transcript timeline traceability, revision-driven re-analysis, and multi-type classroom differentiated analysis. The report clearly distinguishes implemented, partially implemented, and pending functions, objectively explains resource and technical constraints,

and proposes a phased iteration plan. All core functions are verified with real classroom materials, fully proving the practical business value of the project.